

Start 16 Jun 2026	Duration 6 Days	Deadline 21 Jun 2026
AI Layers 5 Models	Team 2 People	Output Live Dashboard

TEAM ROLE LEGEND

AI/ML ENGINEER	Deep Learning specialist · Owns all model training, feature engineering, ML pipeline, planning engine, geo-clustering
MERN + ML DEV	Full-stack MERN developer with ML exposure · Owns Streamlit dashboard, map viz, field PDF, API glue, GitHub, demo
BOTH	Collaborative task · Both engineers contribute together

ROLE RESPONSIBILITY SUMMARY

AI/ML ENGINEER (DL Specialist) • Data loading, cleaning, feature engineering (all 13 features) • KMeans geo-clustering and EDA charts • XGBoost Severity Classifier + Duration Regressor • RandomForest Road Closure Classifier • Optional: DL/MLP layer using PyTorch/TensorFlow (DL specialty) • planner.py — manpower, barricade, equipment, ETA rules engine • diversion.py — route scoring algorithm + junction_db.py • Model evaluation, CV, feature importance plots • Edge case handling in ML pipeline • Model eval tab inside Streamlit dashboard	MERN + ML DEVELOPER • GitHub repo setup, folder structure, requirements.txt • Streamlit app skeleton and all UI layout • Sidebar input form (dropdowns, sliders) • Prediction display cards wired to planner.py • Folium map — base map, incident markers, barricade markers, diversion lines, heatmap • streamlit-folium map embedding • Historical insights panel (st.expander) • Field action sheet PDF (reportlab) • EDA charts tab (Plotly) • README.md, demo video recording, HackerEarth submission
--	---

PHASE 1 · Data Ingestion, Cleaning & EDA

Day 1 · 16 June 2026

AI Focus: No model yet — this phase DEFINES model quality. Bad features = bad AI.

Task	Details & Notes	When	Owner
Load dataset	pd.read_csv(); inspect shape, dtypes, null counts, value counts	16 Jun AM	AI/ML ENG
Fix lat/lng zeros	Replace 0.0 lat/lng with NaN; impute via geopy if needed	16 Jun AM	AI/ML ENG
Parse datetimes	pd.to_datetime() on start/end/closed; extract hour, weekday, month	16 Jun AM	AI/ML ENG
Compute duration	duration_mins = (closed - start).dt.total_seconds()/60 → new column	16 Jun AM	AI/ML ENG
Encode categoricals	event_cause, veh_type, corridor, zone, junction → label encode	16 Jun PM	AI/ML ENG
Flag peak hours	is_peak_hour binary: 1 if 7–10 AM or 5–8 PM	16 Jun PM	AI/ML ENG
KMeans geo-cluster	Cluster lat/lon into k=10 zones; add lat_cluster column	16 Jun PM	AI/ML ENG
EDA charts	Top-5 causes, incidents by hour/day, H/M/L split, corridor heat counts — save as PNG	16 Jun PM	AI/ML ENG
Set up project repo	Init GitHub repo; create folder structure /data /models /app /notebooks; push skeleton	16 Jun PM	MERN DEV
requirements.txt	List all libs: pandas, xgboost, scikit-learn, streamlit, folium, reportlab...	16 Jun PM	MERN DEV
Export clean dataset	Save cleaned DataFrame as clean_incidents.csv; commit to repo	16 Jun EVE	BOTH

Tools: pandas · numpy · matplotlib · seaborn · scikit-learn (KMeans) · geopy**Deliverable:** clean_incidents.csv · eda_insights.ipynb · 5 EDA charts (PNG) · GitHub repo skeleton

PHASE 2 · AI Model Training — 3 Prediction Engines

Day 2 · 17 June 2026

AI Focus: 3 models trained: Severity Classifier + Duration Regressor + Closure Classifier			
Task	Details & Notes	When	Owner
Define X and y	X = all engineered features; y1=priority, y2=duration_mins, y3=requires_closure	17 Jun AM	AI/ML ENG
Train/test split	80/20 split, stratify=y1, random_state=42	17 Jun AM	AI/ML ENG
Model 1 – Severity	XGBoostClassifier(n_estimators=200, max_depth=5) → High/Med/Low · target: Accuracy > 72%	17 Jun AM	AI/ML ENG
Model 2 – Duration	XGBoostRegressor(n_estimators=200) → minutes · target: RMSE < 30 min	17 Jun PM	AI/ML ENG
Model 3 – Road Closure	RandomForestClassifier(n_estimators=150) → Yes/No + prob · target: F1 > 0.70	17 Jun PM	AI/ML ENG
DL enhancement (optional)	Try a lightweight MLP or LSTM on time-series features if XGB underperforms	17 Jun PM	AI/ML ENG
Feature importance plots	Top-15 features per model; save as PNG for dashboard EDA tab	17 Jun PM	AI/ML ENG
Confusion matrix + eval	Precision/recall/F1 for classifiers; log to model_eval_report.txt	17 Jun PM	AI/ML ENG
5-fold cross-validation	CV on each model; log mean ± std to results.txt	17 Jun EVE	AI/ML ENG
Save all models	joblib.dump → model1.pkl, model2.pkl, model3.pkl; test reload works; commit .pkl files	17 Jun EVE	AI/ML ENG
Wire model outputs to app	MERN dev reads model_eval_report.txt; drafts prediction display cards in Streamlit skeleton	17 Jun EVE	MERN DEV
Tools: XGBoost · scikit-learn · joblib · matplotlib · pandas · numpy Deliverable: model1.pkl · model2.pkl · model3.pkl · model_eval_report.txt · Streamlit card skeleton			

PHASE 3 · AI Planning Engine — Manpower, Barricades & Equipment		Day 3 · 18 June 2026	
AI Focus: Core differentiator: AI outputs field-ready orders — not just a score			
Task	Details & Notes	When	Owner
Design planning schema	Output dict: {manpower, barricade_points, equipment, urgency_level, eta}	18 Jun AM	AI/ML ENG
Manpower rules matrix	severity × road_closure × duration → officer count (see table in original plan)	18 Jun AM	AI/ML ENG
Barricade placement logic	closure=True → +4 pts; junction=True → +2 pts; high → +3 pts	18 Jun AM	AI/ML ENG
Equipment list generator	breakdown→tow+cones; tree→saw+cones; accident→ambulance	18 Jun AM	AI/ML ENG
Urgency & ETA logic	High+closure→8 min; High+no closure→12 min; Low→20 min	18 Jun PM	AI/ML ENG
Integrate models → plan	Full pipeline: raw input → features → 3 models → planning engine → output dict	18 Jun PM	AI/ML ENG
Historical context lookup	'Last 3 similar incidents on corridor averaged X min' — get_history() fn	18 Jun PM	AI/ML ENG
Export as planner.py	Clean module with generate_plan(input_dict) API; tested and importable	18 Jun EVE	AI/ML ENG
Streamlit input form	Sidebar: dropdowns (cause, corridor, zone, junction, veh_type); sliders (hour, day)	18 Jun AM	MERN DEV
Connect planner to UI	Call planner.generate_plan() from Streamlit; display severity badge, duration bar, closure %	18 Jun PM	MERN DEV
Deployment plan cards	Styled metric cards: Officers needed Barricade points Equipment list with icons	18 Jun EVE	MERN DEV
Tools: Python (rules engine) · pandas · joblib · streamlit Deliverable: planner.py · Streamlit input form + prediction cards live			

PHASE 4 · Diversion Route Engine + Map Visualization			Day 4 · 19 June 2026
AI Focus: Route scoring algorithm ranks alternates by distance + historical congestion + time-of-day			
Task	Details & Notes	When	Owner

Build junction lookup DB	Hardcode 15+ Bengaluru junctions → lat/lon + alternate corridors in junction_db.py	19 Jun AM	AI/ML ENG
Route scoring function	Score = distance penalty + congestion history + time slot; score_route() fn	19 Jun AM	AI/ML ENG
Top-3 diversion recommender	For any junction → return 3 scored alternate routes with reason; get_diversions()	19 Jun AM	AI/ML ENG
Folium base map	Bengaluru center (lat 12.97, lon 77.59) with OpenStreetMap tile layer	19 Jun AM	MERN DEV
Incident markers	Color pins: Red=High, Orange=Med, Green=Low; click = popup with details	19 Jun PM	MERN DEV
Barricade point markers	Blue diamond markers at recommended barricade coordinates	19 Jun PM	MERN DEV
Diversion route lines	PolyLine from incident → diversion corridor; dashed yellow line style	19 Jun PM	MERN DEV
Historical heatmap layer	HeatMap plugin showing past incident density across Bengaluru; toggle layer	19 Jun PM	MERN DEV
Export map HTML	map.save('bengaluru_traffic_map.html'); test embeds cleanly in Streamlit	19 Jun EVE	MERN DEV
Wire diversion to planner	Connect get_diversions() output to map + ranked route list in Streamlit	19 Jun EVE	BOTH

Tools: folium · folium.plugins · branca · geopy · streamlit-folium

Deliverable: diversion.py · junction_db.py · bengaluru_traffic_map.html · Streamlit map tab live

PHASE 5 · Streamlit AI Dashboard — Full Integration

Day 5 · 20 June 2026

AI Focus: Demo flow: Input event → AI predicts → Plans deployment → Shows map → Downloads field PDF

Task	Details & Notes	When	Owner
App skeleton	Streamlit layout: sidebar inputs main panel map panel field sheet tab	20 Jun AM	MERN DEV
Prediction display	Call planner.generate_plan() → severity badge, duration bar, closure % with color coding	20 Jun AM	MERN DEV
Deployment plan cards	Metric cards: Officers needed Barricade points Equipment list with icons	20 Jun PM	MERN DEV
Diversion route display	Top-3 alternate routes as ranked list with distance + ETA added	20 Jun PM	MERN DEV
Embed Folium map	streamlit_folium.folium_static(map) — incident pin + barricades + diversions	20 Jun PM	MERN DEV
Historical insights panel	st.expander: similar past incidents, avg duration, peak times for that zone	20 Jun PM	MERN DEV
Field action sheet PDF	Generate printable PDF via reportlab — officer count, positions, routes + PDF download button	20 Jun EVE	MERN DEV
EDA charts tab	Show pre-generated EDA PNGs: incident heatmap, cause breakdown, peak hours via Plotly	20 Jun EVE	MERN DEV
DL model integration	If DL model trained Day 2, wire it as secondary severity predictor; compare output	20 Jun PM	AI/ML ENG
Model performance tab	Add model eval report + feature importance charts to dashboard 'About AI' tab	20 Jun EVE	AI/ML ENG
End-to-end smoke test	Both run full flow together: enter input → check all panels render correctly	20 Jun EVE	BOTH

Tools: streamlit · streamlit-folium · plotly · pandas · joblib · reportlab

Deliverable: app.py — full Streamlit dashboard, locally runnable with one command

PHASE 6 · Test, Polish & Submit

Day 6 · 21 June 2026

AI Focus: Final push — make it look like a product, not a notebook

Task	Details & Notes	When	Owner
End-to-end test (5 cases)	High breakdown Tumkur AM; Low tree fall night; Med accident ORR; edge cases	21 Jun AM	BOTH
Edge case handling	Unknown junction → fallback; zero duration → skip; missing zone → default	21 Jun AM	AI/ML ENG
UI polish	App title, logo placeholder, color theme, meaningful metric labels	21 Jun AM	MERN DEV

Write README.md	Sections: Problem · Dataset · Approach · AI Models · How to Run · Tech Stack · Screenshots	21 Jun PM	MERN DEV
GitHub push	Push all: notebooks, .py files, models (.pkl), requirements.txt; repo public + runnable	21 Jun PM	BOTH
Record demo video	2–3 min: problem → input → AI plan → map → field PDF. Practice demo script below.	21 Jun PM	MERN DEV
Final model check	Re-run CV scores; confirm no data leakage; attach eval report to submission	21 Jun PM	AI/ML ENG
Submit on HackerEarth	Upload code link + demo video + README to submission portal	21 Jun EVE	BOTH

Tools: GitHub · Loom / OBS · HackerEarth portal

Deliverable: Public GitHub repo · Demo video (2–3 min) · HackerEarth submission confirmed ✓

6-DAY DAILY CHECKLIST

Date	Day	AI/ML Engineer Must Achieve	MERN+ML Dev Must Achieve
16 Jun	Day 1	Clean dataset · 13 features · 5 EDA charts saved	Repo skeleton · requirements.txt on GitHub
17 Jun	Day 2	3 ML models trained, evaluated & saved as .pkl	Streamlit card skeleton wired to eval report
18 Jun	Day 3	planner.py works end-to-end for any input	Input form live · deployment plan cards render
19 Jun	Day 4	diversion.py + junction_db.py complete	Folium map renders · all layers togglable
20 Jun	Day 5	DL model wired · model eval tab in dashboard	Full dashboard live · field PDF download works
21 Jun	Day 6	Edge cases handled · CV scores attached	README done · demo video recorded · submitted ✓

COMPLETE TECH STACK + OWNER

Layer	Library/Tool	Purpose	Owner
Data	pandas + numpy	Load, clean, engineer features	AI
Geo Clustering	scikit-learn KMeans	Cluster lat/lon into hotspot zones	AI
Geocoding	geopy (Nominatim)	Convert addresses → lat/lon	AI
AI Model 1	XGBoost Classifier	Predict incident severity (H/M/L)	AI
AI Model 2	XGBoost Regressor	Predict incident duration in minutes	AI
AI Model 3	RandomForest Classifier	Predict road closure probability	AI
DL Optional	TensorFlow / PyTorch MLP	Deep learning severity enhancer	AI
Model Persistence	joblib	Save & load trained .pkl model files	AI
Map Visualization	folium + plugins	Interactive incident + diversion map	MERN
Dashboard	Streamlit	Full web UI — inputs, charts, map	MERN
Map in Streamlit	streamlit-folium	Embed Folium map in Streamlit app	MERN
Charts	plotly express	Interactive EDA charts in dashboard	MERN
Field PDF	reportlab	Generate printable field action sheet	MERN
Templating	Jinja2 (optional)	Field report template rendering	MERN
Version Control	Git + GitHub	Code repository + submission	BOTH
Demo Recording	Loom / OBS Studio	Screen record 2–3 min demo video	MERN

HACKATHON WINNING TIPS

Tip	Guidance	Owner
MVP First	Get rough end-to-end flow working before polishing. Judges want to see it work — 70% accuracy is fine.	BOTH
Map = Wow Factor	Folium map with colored pins, barricade markers, dashed diversion lines will impress far more than +0.01 accuracy.	MERN DEV
Field Sheet = Unique	No other team will output a printable field action sheet for police. MERN dev owns this — it's your biggest differentiator.	MERN DEV

Show Numbers	Always display: Officers: 20 Barricades: 10 ETA: 8 min Duration: 47 min. Specific numbers make AI feel real.	AI/ML ENG
DL as Bonus	If time allows, AI engineer adds a small MLP to compare vs XGBoost — mention it in the demo as 'deep learning layer'.	AI/ML ENG
Demo Script	'Incident at Jalahalli Cross, 8AM Monday. Our AI predicts HIGH severity, 47-min duration. 20 officers, 10 barricades, reroutes via Peenya.'	BOTH
Clean GitHub	README with screenshots, requirements.txt, folder structure: /data /models /app /notebooks — MERN dev owns this.	MERN DEV
Real Junctions	Name actual Bengaluru junctions in your demo. Makes the system feel deployed, not academic.	BOTH